



Online social activity time predicts ADHD problems in youth from late childhood to early adolescence in the ABCD study

Huiying Deng¹ · Kunru Song¹ · Xiaomin Geng¹ · Linxuan Xu¹ · Jialin Zhang¹ · Xin Li¹ · Jinbo He² · Marc N. Potenza^{3,4,5,6,7,8} · Jintao Zhang¹

Received: 7 April 2024 / Accepted: 19 November 2024 / Published online: 26 December 2024
© Springer-Verlag GmbH Germany, part of Springer Nature 2024

Abstract

Online social interactions increase into adolescence. Although cross-sectional studies have positively associated online social activity (OSA) time and attention-deficit/hyperactivity disorder (ADHD) problems, the directionality remains unclear. Therefore, we examined longitudinal associations between OSA time and ADHD problems using data from the Adolescent Brain Cognitive Development (ABCD) study. Four waves of ABCD data from 11,819 youth participants (52.1% boys; Baseline: $M_{age} = 9.92$ years, $SD_{age} = 0.62$) were utilized. Random-intercepts cross-lagged panel models (RI-CLPMs) adjusting for potential confounds were employed to estimate longitudinal bidirectional associations. Multiple-group RI-CLPM analyses examined potential moderation effects of OSA content type and child's sex. The results revealed that greater OSA time was associated with more ADHD problems in early adolescence, while more ADHD problems did not predict greater OSA time. In moderation analyses, the cross-lagged effects of OSA time on ADHD problems were only significant for girls, and boys showed an inverse relation between ADHD problems and subsequent OSA time. No moderation effect of OSA content type was observed. These findings highlight the potential detrimental effect of OSA time on the development of attentional processes, especially for girls, thereby offering insights that could guide the development and targeting of interventions to mitigate future risks for ADHD problems during adolescence.

Keywords Online social activity time · ADHD problems · Longitudinal studies · Social media · Random-intercept cross-lagged panel models

✉ Jinbo He
hjb@ccnu.edu.cn

✉ Jintao Zhang
zhangjintao@bnu.edu.cn

¹ State Key Laboratory of Cognitive Neuroscience and Learning and IDG/McGovern Institute for Brain Research, Beijing Normal University, Beijing 100875, China

² Key Laboratory of Adolescent Cyberpsychology and Behavior (Ministry of Education), School of Psychology, Central China Normal University, Wuhan 430079, China

³ Department of Psychiatry, Yale University School of Medicine, New Haven, CT, USA

⁴ Child Study Center, Yale University School of Medicine, New Haven, CT, USA

⁵ Department of Neuroscience, Yale University, New Haven, CT, USA

⁶ Connecticut Council on Problem Gambling, Wethersfield, CT, USA

⁷ Connecticut Mental Health Center, New Haven, CT, USA

⁸ Wu Tsai Institute, Yale University, New Haven, CT, USA

Introduction

The transition from late childhood into early adolescence (approximately ages 10 to 15) is marked by growing independence, increasing autonomy and identity formation, alongside diminishing parental influence and monitoring [1, 2]. During this period, more limited parent-to-child interactions are gradually replaced by more extensive peer-to-peer interactions, often involving social media [3]. With technological advances, online social platforms allow access to social information often immediately and at all hours, offering children and adolescents unprecedented opportunities for interpersonal connection and identity exploration [4]. A recent survey found that 96% of U.S. teens reported going online daily, and 46% stayed online “almost constantly” [5]. Meanwhile, the brain is still undergoing significant development, making this a sensitive period for social development and self-perception [6, 7]. These biological, psychological,

and cognitive changes arguably render youth particularly susceptible to online social environments [8, 9].

Although online social platforms offer numerous advantages, recent studies have raised concerns that frequent engagement in online social activity (OSA) may lead to increased problems related with attention-deficit/hyperactivity disorder (ADHD) [10, 11]. ADHD problems are characterized by atypical levels of inattention, hyperactivity, and impulsivity, usually having their onset in childhood [12, 13]. However, the direction of the association between engagement in OSA and ADHD problems remains partly unclear, as most studies have investigated cross-sectional correlations, with limited empirical evidence in childhood and adolescence [14, 15].

According to the scan-and-shift hypothesis, the rapid nature of digital media requires individuals to frequently shift attention, hindering the development of attention/focusing skills and resulting in more ADHD problems [16, 17]. Compared with off-line social interactions, OSA is often characterized by a fast-paced and fragmented nature, as individuals often switch back and forth between multiple conversations or rapidly scroll through brief segments of visual information. As a result, youth who extensively participate in OSA may find it challenging to internally regulate their attention after having become used to or reliant on external regulation through media platforms [18]. In line with this view, several studies have found that time spent engaging in OSA (e.g., using social media, video chatting) could positively predict subsequent ADHD problems among adolescents [19, 20]. Nevertheless, such effects may be reciprocal in nature over time [21, 22]. According to the reinforcing spirals model, media use and media effects (i.e., certain behaviors) contribute to a reciprocal over-time influence process in which the media effect promotes its change [23]. The differential susceptibility to media effects model (DSMM) posits that individuals tend to expose themselves to media content that aligns with their dispositions [14]. It has also been suggested that youth with ADHD problems may be more likely to engage in OSA, which allows for simultaneous involvement in multiple conversations and immediate rewards (e.g., “likes”) [24, 25]. Additionally, youth with ADHD problems often have difficulties in establishing or maintaining friendships [26, 27]. From a social compensation perspective [28], youth with attention and social concerns may turn to online social platforms as a more convenient and controllable way to compensate for their limited offline social networks.

Despite the possibility of the reciprocal associations, most previously reported relations between OSA time and ADHD problems were derived from cross-sectional studies. Prior longitudinal studies of this topic have failed to consider bidirectional effects, instead conceptualizing media

use as a possible cause of ADHD problems (e.g., examining the effect of social media use at T1 on ADHD problems at T2) [19, 20]. In fact, only one study has investigated reciprocal relations longitudinally [10]. Specifically, Boer et al. used a three-wave prospective design among 543 Dutch adolescents aged 11–15 years and found that social media use problems were unidirectionally related to increased ADHD problems. However, a recent study has shown that a 3-time-point model is relatively insufficient for modeling within-person change longitudinally and suggested including 4 or more time points for model identification [29]. Furthermore, most existing studies on media use have mainly focused on adolescents, with limited studies exploring these associations during the transition from childhood to adolescence, which is a period of increased sensitivity to OSA and a critical time for cognitive development [8]. Therefore, the primary aim of this study was to estimate the longitudinal bidirectional associations between OSA time and ADHD problems from late childhood to early adolescence in a large and diverse youth sample of the Adolescent Brain and Cognitive Development (ABCD) Study [30].

Beyond disentangling directionality, it is important to know for whom and under what conditions a relation between OSA time and ADHD problems may manifest [31]. Apart from time-based factors, content-based factors may promote a more comprehensive understanding of OSA engagement [32]. Previous research has demonstrated that the type of OSA content (e.g., image vs. text) can have differential impacts on well-being [33, 34]. The DSMM also proposes that the relation between media use and ADHD problems may differ depending on the type of media content [14, 21]. In comparison to image-based platforms (e.g., Instagram), video-based platforms (e.g., TikTok) are characterized by more rapid changes in focus and fast-paced, quickly changing features (e.g., animation, sounds). These features may force individuals to process rapidly intense audiovisual stimuli and shift their attention more frequently. Prolonged usage may thus make it more difficult for individuals to engage in activities requiring sustained attention [35]. Besides usage content, sex is a key dispositional factor often being considered when investigating media effects. Previous research has shown that males and females exhibit different tendencies for online activity options and usage patterns [9, 36]. Specifically, compared with males, females appear more susceptible to problems with OSA [37]. However, the factors influencing relations between OSA time and ADHD problems remain largely unexplored. Thus, this study also aimed to explore potential influencing factors, focusing on potential effects of OSA content type and sex.

This study aimed to investigate longitudinal relations between OSA time and ADHD problems, utilizing four waves of data collected annually over four years from the

ABCD study. By applying random-intercepts cross-lagged panel models (RI-CLPMs), we aimed to separate time-varying within-person effects from stable between-person effects, thereby allowing for more accurate estimations of directionality [38]. The current study was pre-registered at <https://doi.org/10.17605/OSF.IO/KXYZP>. Based on existing literature, we hypothesized a reciprocal longitudinal relation between OSA time and ADHD problems over time. Specifically, OSA time would relate to subsequent within-person increases in ADHD problems, and ADHD problems would relate to subsequent within-person increases in OSA time. In exploratory analyses, we explored whether OSA content type and sex would moderate bidirectional associations.

Methods

Participants

The ABCD Study is a multisite, longitudinal study with 11,868 participants enrolled at baseline from 21 U.S. sites. Data analyzed were from the ABCD 5.0 release, which includes the baseline (T1: 2016–2018, ages 9 and 10 years), 1-year follow-up (T2: 2017–2019, ages 10 and 11 years), 2-year follow-up (T3: 2018–2020, ages 11 and 12 years) and 3-year follow-up (T4: 2019–2021, ages 12 and 13 years) assessments. Participants with missing data for baseline OSA time ($N=43$) or ADHD problems ($N=8$) were excluded, leaving 11,819 participants for analyses.

Measures

Online social activity (OSA) time

OSA time was measured using the ABCD Youth Screen Time Survey. Youth were asked how much time they spent on six screen activities (watching TV shows or movies, watching videos, playing videogames, texting, visiting social networking sites, and video chatting) for both weekdays and weekends. As previously [39], we summed the time spent on “texting”, “visiting social networking sites” and “video chatting” into a combined category of OSA and calculated the weighted average participants’ typical weekday and weekend OSA times ($((\text{weekday average} \times 5) + (\text{weekend average} \times 2))/7$) to obtain typical week measures.

ADHD problems

ADHD problems (including inattention, hyperactivity, and impulsivity) were assessed by the Child Behavior Checklist (CBCL) on the attention problems syndrome subscale,

to which parents reported on a scale from 0 (not true) to 2 (very true/often true) in response to statements concerning ADHD problems in their children [40–42]. The resulting CBCL-derived T-scores were used for analyses, with higher scores indicating more severe problems.

The most often used online social platforms

Beginning in Year 2, participants who owned at least one online social account were asked which online social platform (e.g., YouTube, Instagram, Snapchat, Facebook) they used the most. By focusing on the content type (i.e., video or image) of each online social platform, we defined two groups of individuals who mostly engaged in video-based and image-based OSA, respectively. Due to a substantial percentage of missing responses to this item in Year 2 (49.3%), we resorted to participants’ responses in Year 3 for the purpose of grouping.

Covariates

Potential sociodemographic covariates for understanding associations between OSA time and ADHD problems were selected based on the existing literature and theories [11, 39, 43]. Age, sex, race, ethnicity, highest level of parental education, family income, and parental marital status were based on parents’/caregivers’ self-reports. Accounting for the potential interplay between OSA time and time spent on the remaining three activities (watching TV shows or movies, watching videos, and playing videogames), and given that these activities have also been found to associate with ADHD problems [44, 45], we collapsed the remaining three activities into a combined category of “video-centric activity” and included it as a covariate. Since the time spent on “watching videos” was not measured in Year 3, we only summed the time spent on “watching TV shows or movies” and “playing videogames” in Year 3.

Statistical analysis

The longitudinal bidirectional associations between OSA time and ADHD problems were examined using RI-CLPMs. We included baseline age, sex, race, ethnicity, parental marital status, highest level of parental education, and family income as time-invariant covariates, and time spent on remaining three non-social activities as a time-variant covariate in the RI-CLPMs. Missing data for baseline age (0.01%), race (1.43%), ethnicity (1.29%), parental marital status (0.81%), highest level of parental education (0.12%) and family income (8.57%) were imputed using multiple imputation by chained equations with the “mice” package (v3.16.0) in R Statistical Software (v4.3.1) [46,

47]. We generated 100 imputed datasets, and the final dataset was derived by combining imputations using modes and medians for categorical and continuous variables, respectively. The supplementary materials also present the results obtained using the non-imputed dataset. Moderating effects of OSA content type and sex were further tested by comparing a multigroup model that specified no constraints between groups with a multigroup model in which the cross-lagged paths were constrained to be equal between groups. A Satorra-Bentler chi-square test was

performed to test whether this constraint could be imposed without reducing the model adjustment. If the test result was significant, potential group differences were assumed to be present. For all RI-CLPMs, the maximum likelihood robust (MLR) estimator was used, and missing data were handled with full information maximum likelihood estimation (FIML). Model fit was evaluated by the comparative fit index (CFI), Tucker–Lewis index (TLI), root mean square error of approximation (RMSEA), and standardized root mean squared residual (SRMR) and interpreted using common thresholds of good fit [48].

Table 1 Baseline sociodemographic and study characteristics of ABCD Study participants included in the analytic sample

Characteristic	Mean (SD)/%
Age (years)	9.92 (0.62)
Sex, n (%)	
Male	6163 (52.1%)
Female	5656 (47.9%)
Race	
White	7587 (64.2%)
Black	1865 (15.8%)
Asian	273 (2.3%)
Mixed/Other	2094 (17.7%)
Parental marital status	
Single	3137 (26.5%)
Married or living with partner	8682 (73.5%)
Ethnicity	
Non-Hispanic	9427 (79.8%)
Hispanic	2392 (20.2%)
Highest level of parental education	
High school or less	3221 (27.3%)
College education	8598 (72.7%)
Family income	
<\$25,000	2124 (18.0%)
\$25,000–\$49,999	1635 (13.8%)
\$50,000–\$74,999	1505 (12.7%)
\$75,000–\$99,999	1678 (14.2%)
\$100,000–\$199,999	3630 (30.7%)
\$200,000+	1247 (10.6%)
Screen time variables	
OSA time at T1	0.54 (1.15)
OSA time at T2	0.89 (1.57)
OSA time at T3	1.58 (2.15)
OSA time at T4	2.44 (2.63)
VCA time at T1	3.26 (2.48)
VCA time at T2	3.60 (2.62)
VCA time at T3	4.03 (2.71)
VCA time at T4	3.77 (2.17)
ADHD problems T scores	
ADHD problems at T1	53.90 (6.17)
ADHD problems at T2	53.71 (5.98)
ADHD problems at T3	53.47 (5.58)
ADHD problems at T4	53.61 (5.62)

Note. OSA=online social activity; VCA=video-centric activity; T1=Baseline; T2=1-year follow-up; T3=2-year follow-up; T4=3-year follow-up

Results

Descriptive statistics

Table 1 describes baseline sociodemographic characteristics of the 11,819 participants included. Baseline differences in sociodemographic variables between the final sample and excluded cases due to missing data were also tested (Table S1). Cross-sectional correlations between OSA time and ADHD problems and descriptive statistics at the 4 waves are presented in Table 2. Higher levels of OSA time were significantly associated with higher levels of ADHD problems at each wave.

Regarding online social platforms most frequently used at the 3-year follow-up, TikTok (35.9%) and YouTube (21.7%) were used the most, followed by Instagram (16.7%) and Snapchat (13.7%). After those platforms came Multiplayer Videogame Online Chatting Platforms with 5.7% and smaller shares who mostly used Pinterest, Reddit, Facebook, Twitter, and Tumblr (Figure S1). Participants who answered they most used “TikTok”, “YouTube” and “Multiplayer Videogame Online Chatting Platforms” were classified as having video-based OSA use, and the remaining participants were classified as having image-based use, except for participants who answered “Other”.

Random-intercepts, cross-lagged panel model

The overall model fit of the RI-CLPM was good (CFI=0.953; TLI=0.920; RMSEA=0.040; SRMR=0.044; $\chi^2(102)=2028.860$, $p<0.001$). In the between-person part, there were significant variances (i.e., random intercepts) for both OSA ($\sigma^2=0.82$; 95% CI, 0.68–0.95) and ADHD problems ($\sigma^2=15.18$; 95% CI, 13.75–16.61), revealing individual differences in the person-level means of both variables. In addition, a positive correlation between the random intercepts ($r=0.068^{**}$; 95% CI, 0.07–0.35) suggests that children with higher levels of OSA time also exhibit more ADHD problems on average, and across all study waves.

Table 2 Bivariate correlation coefficients between OSA time and ADHD problems as well as descriptive statistics

	1	2	3	4	5	6	7	8
1. OSA time T1	-							
2. OSA time T2	0.400***	-						
3. OSA time T3	0.315***	0.459***	-					
4. OSA time T4	0.271***	0.404***	0.556***	-				
5. ADHD T1	0.061***	0.038***	0.011	0.008	-			
6. ADHD T2	0.057***	0.032***	0.016	0.010	0.729***	-		
7. ADHD T3	0.056***	0.049***	0.040***	0.025*	0.683***	0.725***	-	
8. ADHD T4	0.060***	0.041***	0.033***	0.046***	0.618***	0.666***	0.722***	-
N	11,819	11,157	10,884	10,259	11,819	11,165	8053	7428

Note. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$

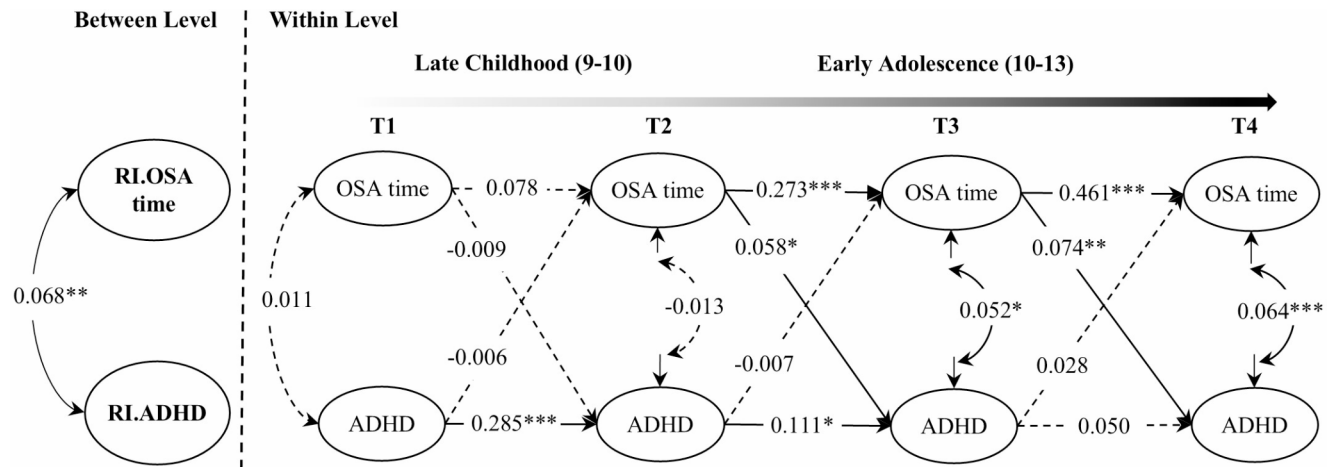


Fig. 1 Random intercept cross-lagged panel model of OSA time and ADHD problems. All analyses adjusted for time-invariant and time-varying covariates. To minimize complexity, the covariates included

are not shown. All coefficients are standardized; significant paths are shown using solid lines. OSA = online social activity; ADHD = ADHD problems. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$

In the within-person (or time-variant) component of the model (Fig. 1; Table S2; Table S3), positive autocorrelations suggest that children's longitudinal trajectories for OSA time and ADHD problems were relatively stable across adjacent time points. Children's OSA time at baseline was not associated with OSA time at 1-year follow-up, which signals a lack of carryover effect for OSA time in late childhood. Longitudinal trajectories for OSA time began to stabilize between the ages of 10 and 13 years, with very high levels of stabilization in early adolescence. ADHD problems were initially stable between the ages of 9 and 12. However, a subsequent nonsignificant autoregressive parameter indicates a lack of carryover of children's ADHD problems between the ages of 12 and 13.

Adolescents with high levels of OSA time did not exhibit significantly increased ADHD problems one year later until between the ages of 10 and 13. The obverse direction of more ADHD problems being associated with higher levels of OSA time was not observed. Taken together, these findings suggest that higher levels of OSA time were associated with significantly more ADHD problems subsequently in early adolescence but not vice versa.

Potential moderating roles of OSA content and sex

No moderation effect of OSA content type was found, as constraining the cross-lagged effects to be equal across two content types (video vs. image) did not worsen the model fits (Table S5). Considering that alternate classifications of OSA content type may lead to different findings, supplementary analyses with alternate classifications were conducted, yielding similarly nonsignificant results (Table S6, S7). Regarding sex, the cross-lagged effects of OSA time (at T2 and T3) on ADHD problems (at T3 and T4, respectively) were only significant for girls. Furthermore, the cross-lagged parameter linking ADHD problems (at T1 and T2) and OSA time (at T2 and T3, respectively) differed significantly between males and females (Table 3, S8, S9). Specifically, this effect was significantly negative in males, but was not significant in females.

Table 3 Cross-lagged parameter estimates from RI-CLPMs for females and males

Cross Lag	Females (<i>N</i> =5656)			Males (<i>N</i> =6163)		
	β (SE)	95% CI	<i>p</i>	β (SE)	95% CI	<i>p</i>
OSA time → ADHD problems						
T1 → T2	-0.006 (0.033)	[-0.071, 0.059]	0.857	-0.013 (0.023)	[-0.253, 0.141]	0.576
T2 → T3	0.094 (0.041)	[0.013, 0.175]	0.023	-0.001 (0.030)	[-0.139, 0.133]	0.967
T3 → T4	0.080 (0.028)	[0.024, 0.135]	0.005	0.007 (0.036)	[-0.113, 0.138]	0.844
ADHD problems → OSA time						
T1 → T2	0.046 (0.029)	[-0.012, 0.104]	0.117	-0.057 (0.024)	[-0.029, -0.002]	0.020
T2 → T3	0.024 (0.022)	[-0.019, 0.066]	0.281	-0.051(0.021)	[-0.040, -0.004]	0.017
T3 → T4	0.023 (0.027)	[-0.030, 0.076]	0.403	0.025 (0.029)	[-0.025, 0.064]	0.380

Notes. β =standardized coefficients; SE=standard error; CI=95% confidence interval. **Bold** represents estimates for which the 95% CI does not overlap zero ($p < 0.05$)

Discussion

In a large and demographically diverse sample of U.S. children, we examined the bidirectional, within-person relations between OSA time and ADHD problems using RI-CLPMs to separate between- and within-person effects. OSA content type and children's sex were explored as potential moderators of the within-person cross-lagged associations. The findings support a temporal relation between OSA time and ADHD problems from late childhood to early adolescence and may inform the development and targeting of interventions.

Results from the temporally stable between-person components of the RI-CLPM confirmed a positive between-person relation between OSA time and ADHD problems. This indicated that adolescents who reported more OSA time also have more ADHD problems reported by parents, which was in line with previous cross-sectional studies and meta-analyses [11, 16]. The within-person part of RI-CLPM showed that greater OSA time was associated with more ADHD problems one year later, supporting the scan-and-shift hypothesis [17]. When engaging in OSA, youth can converse with dozens of peers simultaneously via text messages or social media. For youth spending more time on OSA, they often shift their attention more frequently when using various online social platforms, and this may result in more ADHD problems. Of note, this positive within-person effect did not emerge until early adolescence (from T2 to T4, aged 10–13), as we found no predictive role of OSA time on subsequent ADHD problems during late childhood (from T1 to T2, aged 9–10). This finding is consistent with a recent study highlighting the more influential role that social media may play at approximately 11–15 years old [9], possibly attributable to maturational changes occurring in this specific period. In addition, it is plausible that OSA time affects children's ADHD problems through a cumulative or additive process. When participants enter early adolescence, their frequency of switching between OSAs may further increase to a level that contributes to the development of

ADHD problems [18]. This further emphasizes the importance of considering developmental stage when investigating the relation between OSA time and ADHD problems, as the association may change during the transition from late childhood to early adolescence [3]. It is noteworthy that youth typically start engaging extensively in OSA around the age of 13 [49], which nearly coincides with the age of onset criterion for ADHD (i.e., before 12 years old) in the DSM-5 [50]. Although ADHD was not diagnosed in this study, future research should consider the possibility that increased OSA time may contribute to ADHD in adolescents even after the threshold of the ADHD age-of-onset criterion. However, the developmental sensitivity window of the relation between OSA time and ADHD problems needs to be further investigated using more time waves.

Contrary to our expectations, the within-person effect of ADHD problems on OSA time was not significant across all waves. Alternatively, the time lags for ADHD problems predicting OSA time might differ from those for the effect of OSA time on ADHD problems [23]. For example, ADHD problems may have an immediate effect on engagement in OSA, while OSA may need longer time to exert influences on ADHD problems. The fixed time lag in the current study did not account for this variability, possibly resulting in an underestimation of the effect of ADHD problems on OSA time. Furthermore, the within-person-oriented study design of the RI-CLPM does not eliminate the possibility that stable levels of ADHD problems at the between-person level (e.g., genetically determined) affected engagement in OSA over time [10].

Based on participants' most used online social platform, we explored whether OSA content would moderate the "OSA time-ADHD problems" association. However, no moderation effect of OSA content type was observed, suggesting that there were no differences in the within-person cross-lagged effects between participants with video-based and image-based OSA use. As youth today do not typically dedicate themselves to just one online social platform, the

boundaries between image and video formats may become too blurred to show differences.

Testing of sex-related differences showed that the within-person effects of OSA time (at T2 and T3) on ADHD problems (at T3 and T4, respectively) were only significant for girls. This suggests that girls with more OSA time may be at greater risk for developing later ADHD problems compared to boys. Studies have indicated that females engage in media multitasking with OSA (i.e., the simultaneous engagement in two or more online social activities) generally more frequently than males, which may be especially detrimental to their ability to focus and maintain attention [18]. Additionally, increases in sensitivity to online social environments typically occur earlier in females than in males [9]. Therefore, the positive predictive effect of OSA time on subsequent ADHD problems may manifest earlier in females. Moreover, the within-person effects of ADHD problems (at T1 and T2) on OSA time one year later (at T2 and T3, respectively) were significantly negative in males and were positive but not significant in females. This suggests that males with ADHD problems may engage in OSA to a lesser degree. These gender-related differences could result from boys and girls having different preferences or motivations when choosing screen activities [36, 51]. Boys with ADHD problems may be less socially oriented and more attracted to other screen activities, such as video gaming (which may be used for socializing), rather than OSA [52].

Limitations and future directions

Study limitations warrant mention. First, OSA time was assessed based on self-report, which may be influenced by social desirability and recall biases. Additionally, time spent on OSA partly reflects participants' involvement in OSA. Future studies combining objective measures (e.g., log-based data obtained through apps) with more comprehensive measures of OSA (e.g., addiction tendencies, use motives, specific content consumed) are indicated. Second, the classification of OSA content type in this study was roughly based on one item, and within-person changes in OSA content type were not considered. Although no moderating effect of OSA content type was observed, future studies should use more valid measures of content-based variables, and could identify individuals who consistently use certain types of OSA to enhance the reliability of the grouping results. Third, how best to classify predominantly image-based versus video-based use may be debated. For example, in 2020, Instagram introduced a Reels component similar to videos found in TikTok, and Snapchat also has video-sending capabilities. Both Instagram and Snapchat being counted as “image based” may explain the non-significant findings. However, compared to TikTok and YouTube, Instagram and Snapchat

are more image-based platforms [34, 53]. Although Instagram posts include both photos and videos, photos constitute a significantly higher proportion (71.9%) compared to videos (14.2%) [54]. Alternate classifications may have led to different findings, although in the case of switching classifications of Instagram and Snapchat to video-based, moderation analyses were similarly negative (see supplemental materials). Fourth, time spent on offline social activities were not measured. Considering possible effects of both online and offline social activities on ADHD problems may provide a more comprehensive understanding, consistent with models for promoting healthy use of the internet [55]. Fifth, due to the use of long time-intervals, potential relations between daily fluctuations in ADHD problems and OSA time might not be observed. Longitudinal research investigating OSA time and ADHD problems using more waves and shorter time-intervals (e.g., experience sampling methods) is needed. Finally, interpretation should be limited to the specific age range of late childhood and early adolescence and US jurisdictions. Further research using more waves is needed to explore whether this unidirectional relation could be generalized to subsequent adolescent time points, and studies of data from other countries are needed to increase generalizability.

Conclusion

The findings of this longitudinal study provide further evidence that greater OSA time related longitudinally to more ADHD problems in early adolescence. This relation was unidirectional, as the reverse pattern was not observed. The present findings should increase awareness regarding OSA time and how it may promote ADHD problems in youth. Early interventions targeting high-frequency OSA may help reduce the risk for developing ADHD problems. Future studies are needed to further investigate this relation at later developmental stages and to understand underlying mechanisms.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00787-024-02620-6>.

Author contributions All authors contributed to the study conception and design. The first draft of the manuscript was written by H.Y.D. and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding This study was supported by the National Natural Science Foundation of China [grant number 32171083 & 32371142, STI 2030–Major Projects 2021ZD0200500], and 111 Project [BP0719032]. The ABCD Study® was supported by the National Institutes of Health and additional federal partners under award numbers U01DA041048, U01DA050989, U01DA051016, U01DA041022, U01DA051018,

U01DA051037, U01DA050987, U01DA041174, U01DA041106, U01DA041117, U01DA041028, U01DA041134, U01DA050988, U01DA051039, U01DA041156, U01DA041025, U01DA041120, U01DA051038, U01DA041148, U01DA041093, U01DA041089, U24DA041123, U24DA041147. A full list of supporters is available at <https://abcdstudy.org/federal-partners.html>. A listing of participating sites and a complete listing of the study investigators can be found at https://abcdstudy.org/consortium_members/. ABCD consortium investigators designed and implemented the study and/or provided data but did not necessarily participate in the analysis or writing of this report. This manuscript reflects the views of the authors and does not reflect the views of the NIH or ABCD consortium investigators.

Data availability The datasets generated and/or analyzed during the current study are available in the ABCD Data repository: <https://nda.nih.gov/abcd/>.

Code availability Code is available upon request.

Declarations

Ethical approval Centralized institutional review board (IRB) approval was obtained from the University of California, San Diego. Study sites obtained approval from their respective IRBs.

Informed consent Written informed consent from parents and assent from children were obtained at each site.

Consent for publication Consent for publication was obtained from all authors.

Competing interests The authors declare no competing interests.

References

- Blakemore SJ, Mills KL (2014) Is adolescence a sensitive period for sociocultural processing? *Annu Rev Psychol* 65:187–207. <https://doi.org/10.1146/annurev-psych-010213-115202>
- Sawyer SM, Azzopardi PS, Wickremarathne D, Patton GC (2018) The age of adolescence. *Lancet Child Adolesc Health* 2:223–228. [https://doi.org/10.1016/S2352-4642\(18\)30022-1](https://doi.org/10.1016/S2352-4642(18)30022-1)
- Madigan S, Reich SM (2023) Consideration of developmental stage and the debate on the effects of screens use-not all things are created equal. *JAMA Pediatr* 177:1123–1124. <https://doi.org/10.1001/jamapediatrics.2023.3670>
- Nesi J, Choukas-Bradley S, Prinstein MJ (2018) Transformation of adolescent peer relations in the social media context: part 1-A theoretical framework and application to dyadic peer relationships. *Clin Child Fam Psychol Rev* 21:267–294. <https://doi.org/10.1007/s10567-018-0261-x>
- Pew Research Center (2023) Teens, Social Media and Technology 2023. <https://www.pewresearch.org/internet/2023/12/11/teens-social-media-and-technology-2023/>. [Accessed December 11, 2023]
- Pfeifer JH, Allen NB (2021) Puberty initiates cascading relationships between neurodevelopmental, social, and internalizing processes across adolescence. *Biol Psychiatry* 89:99–108. <https://doi.org/10.1016/j.biopsych.2020.09.002>
- Crone EA, Konijn EA (2018) Media use and brain development during adolescence. *Nat Commun* 9:588. <https://doi.org/10.1038/s41467-018-03126-x>
- Orben A, Blakemore SJ (2023) How social media affects teen mental health: a missing link. *Nature* 614:410–412. <https://doi.org/10.1038/d41586-023-00402-9>
- Orben A, Przybylski AK, Blakemore SJ, Kievit RA (2022) Windows of developmental sensitivity to social media. *Nat Commun* 13:1649. <https://doi.org/10.1038/s41467-022-29296-3>
- Boer M, Stevens G, Finkenauer C, van den Eijnden R (2020) Attention deficit hyperactivity disorder-symptoms, social media use intensity, and social media use problems in adolescents: investigating directionality. *Child Dev* 91:e853–e865. <https://doi.org/10.1111/cdev.13334>
- Ra CK, Cho J, Stone MD, De La Cerda J, Goldenson NI, Moroney E, Tung I, Lee SS, Leventhal AM (2018) Association of digital media use with subsequent symptoms of attention-deficit/hyperactivity disorder among adolescents. *JAMA* 320:255–263. <https://doi.org/10.1001/jama.2018.8931>
- Biederman J, Mick E, Faraone SV (2000) Age-dependent decline of symptoms of attention deficit hyperactivity disorder: impact of remission definition and symptom type. *Am J Psychiatry* 157:816–818. <https://doi.org/10.1176/appi.ajp.157.5.816>
- American Psychiatric Association (2013) Diagnostic and Statistical Manual of Mental Disorders. American Psychiatric Association. <https://doi.org/10.1176/appi.books.9780890425596>
- Beyens I, Valkenburg PM, Piotrowski JT (2018) Screen media use and ADHD-related behaviors: four decades of research. *Proc Natl Acad Sci* 115:9875–9881. <https://doi.org/10.1073/pnas.1611611114>
- Thorell LB, Burén J, Ström Wiman J, Sandberg D, Nutley SB (2022) Longitudinal associations between digital media use and ADHD symptoms in children and adolescents: a systematic literature review. *Eur Child Adolesc Psychiatry Adv Online Publication*. <https://doi.org/10.1007/s00787-022-02130-3>
- Nikkelen SW, Valkenburg PM, Huizinga M, Bushman BJ (2014) Media use and ADHD-related behaviors in children and adolescents: a meta-analysis. *Dev Psychol* 50:2228–2241. <https://doi.org/10.1037/a0037318>
- Jensen PS, Mrazek D, Knapp PK, Steinberg L, Pfeffer C, Schowalter J, Shapiro T (1997) Evolution and revolution in child psychiatry: ADHD as a disorder of adaptation. *J Am Acad Child Adolesc Psychiatry* 36:1672–1681. <https://doi.org/10.1097/00004583-199712000-00015>
- Baumgartner SE, van der Schuur WA, Lemmens JS, te Poel F (2018) The relationship between media multitasking and attention problems in adolescents: results of two longitudinal studies. *Hum Commun Res* 44:3–30. <https://doi.org/10.1093/hcre.12111>
- McNamee P, Mendolia S, Yerokhin O (2021) Social media use and emotional and behavioural outcomes in adolescence: evidence from British longitudinal data. *Econ Hum Biol* 41:100992. <https://doi.org/10.1016/j.ehb.2021.100992>
- Wallace J, Boers E, Ouellet J, Afzali MH, Conrod P (2023) Screen time, impulsivity, neuropsychological functions and their relationship to growth in adolescent attention-deficit/hyperactivity disorder symptoms. *Sci Rep* 13:18108. <https://doi.org/10.1038/s41598-023-44105-7>
- Valkenburg PM, Peter J (2013) The differential susceptibility to media effects model. *J Commun* 63:221–243. <https://doi.org/10.1111/jcom.12024>
- Slater MD (2015) Reinforcing spirals model: conceptualizing the relationship between media content exposure and the development and maintenance of attitudes. *Media Psychol* 18:370–395. <https://doi.org/10.1080/15213269.2014.897236>
- Slater MD (2007) Reinforcing spirals: the mutual influence of media selectivity and media effects and their impact on individual behavior and social identity. *Communication Theory* 17:281–303. <https://doi.org/10.1111/j.1468-2885.2007.00296.x>

24. Maza MT, Fox KA, Kwon SJ, Flannery JE, Lindquist KA, Prinstein MJ, Telzer EH (2023) Association of habitual checking behaviors on social media with longitudinal functional brain development. *JAMA Pediatr* 177:160–167. <https://doi.org/10.1001/jamapediatrics.2022.4924>
25. Marco R, Miranda A, Schlotz W, Melia A, Mulligan A, Müller U, Andreou P, Butler L, Christiansen H, Gabriels I, Medad S, Albrecht B, Uebel H, Asherson P, Banaschewski T, Gill M, Kuntsi J, Mulas F, Oades R, Roeyers H, Steinhausen HC, Rothenberger A, Faraone SV, Sonuga-Barke EJ (2009) Delay and reward choice in ADHD: an experimental test of the role of delay aversion. *Neuropsychology* 23:367–380. <https://doi.org/10.1037/a0014914>
26. Gardner DM, Gerdes AC (2015) A review of peer relationships and friendships in youth with ADHD. *J Atten Disord* 19:844–855. <https://doi.org/10.1177/1087054713501552>
27. Ginapp CM, Greenberg NR, Macdonald-Gagnon G, Angarita GA, Bold KW, Potenza MN (2023) The experiences of adults with ADHD in interpersonal relationships and online communities: a qualitative study. *SSM Qual Res Health* 3:100223. <https://doi.org/10.1016/j.ssmqr.2023.100223>
28. Zywicki J, Danowski J (2008) The faces of facebookers: investigating Social Enhancement and Social Compensation hypotheses; Predicting Facebook™ and Offline Popularity from sociability and Self-Esteem, and mapping the meanings of Popularity with Semantic Networks. *J Computer-Mediated Communication* 14:1–34. <https://doi.org/10.1111/j.1083-6101.2008.01429.x>
29. Parsons S, McCormick EM (2024) Limitations of two time point data for understanding individual differences in longitudinal modeling - what can difference reveal about change? *Dev Cogn Neurosci* 66:101353. <https://doi.org/10.1016/j.dcn.2024.101353>
30. Volkow ND, Koob GF, Croyle RT, Bianchi DW, Gordon JA, Koroshetz WJ, Pérez-Stable EJ, Riley WT, Bloch MH, Conway K, Deeds BG, Dowling GJ, Grant S, Howlett KD, Matochik JA, Morgan GD, Murray MM, Noronha A, Spong CY, Wargo EM, Warren KR, Weiss SRB (2018) The conception of the ABCD study: from substance use to a broad NIH collaboration. *Dev Cogn Neurosci* 32:4–7. <https://doi.org/10.1016/j.dcn.2017.10.002>
31. Whitlock J, Masur PK (2019) Disentangling the association of screen time with developmental outcomes and well-being: problems, challenges, and opportunities. *JAMA Pediatr* 173:1021–1022. <https://doi.org/10.1001/jamapediatrics.2019.3191>
32. Valkenburg PM (2022) Social media use and well-being: what we know and what we need to know. *Curr Opin Psychol* 45:101294. <https://doi.org/10.1016/j.copsyc.2021.12.006>
33. Pittman M, Reich B (2016) Social media and loneliness: why an Instagram picture may be worth more than a thousand Twitter words. *Comput Hum Behav* 62:155–167. <https://doi.org/10.1016/j.chb.2016.03.084>
34. Masciantonio A, Bourguignon D, Bouchat P, Balty M, Rimé B (2021) Don't put all social network sites in one basket: Facebook, Instagram, Twitter, TikTok, and their relations with well-being during the COVID-19 pandemic. *PLoS ONE* 16:e0248384. <https://doi.org/10.1371/journal.pone.0248384>
35. Konok V, Liskai-Peres K, Bunford N, Ferdinandy B, Júranyi Z, Ujfalussy DJ, Réti Z, Pogány Á, Kampis G, Miklósi Á (2021) Mobile use induces local attentional precedence and is associated with limited socio-cognitive skills in preschoolers. *Comput Hum Behav* 120:106758. <https://doi.org/10.1016/j.chb.2021.106758>
36. Su W, Han X, Yu H, Wu Y, Potenza MN (2020) Do men become addicted to internet gaming and women to social media? A meta-analysis examining gender-related differences in specific internet addiction. *Comput Hum Behav* 113:106480. <https://doi.org/10.1016/j.chb.2020.106480>
37. Blomfield Neira CJ, Barber BL (2014) SNS use and adolescent indicators of adjustment. *Australian J Psychol* 66:56–64. <https://doi.org/10.1111/ajpy.12034>
38. Hamaker EL, Kuiper RM, Grasman RP (2015) A critique of the cross-lagged panel model. *Psychol Methods* 20:102–116. <https://doi.org/10.1037/a0038889>
39. Bagot KS, Tomko RL, Marshall AT, Hermann J, Cummins K, Ksinan A, Kakalis M, Breslin F, Lisdahl KM, Mason M, Redhead JN, Squeglia LM, Thompson WK, Wade T, Tapert SF, Fuemmeler BF, Baker FC (2022) Youth screen use in the ABCD® study. *Dev Cogn Neurosci* 57:101150. <https://doi.org/10.1016/j.dcn.2022.101150>
40. Achenbach T, Rescorla L (2001) Manual for the ASEBA school-age forms & profiles: an integrated system of multi-informant assessment. ASEBA, Burlington, VT
41. Lampert TL, Polanczyk G, Tramontina S, Mardini V, Rohde LA (2004) Diagnostic performance of the CBCL-Attention Problem Scale as a screening measure in a sample of Brazilian children with ADHD. *J Atten Disord* 8:63–71. <https://doi.org/10.1177/108705470400800204>
42. Barch DM, Albaugh MD, Avenevoli S, Chang L, Clark DB, Glantz MD, Hudziak JJ, Jernigan TL, Tapert SF, Yurgelun-Todd D, Alia-Klein N, Potter AS, Paulus MP, Prouty D, Zucker RA, Sher KJ (2018) Demographic, physical and mental health assessments in the adolescent brain and cognitive development study: rationale and description. *Dev Cogn Neurosci* 32:55–66. <https://doi.org/10.1016/j.dcn.2017.10.010>
43. Nagata JM, Chu J, Ganson KT, Murray SB, Iyer P, Gabriel KP, Garber AK, Bibbins-Domingo K, Baker FC (2023) Contemporary screen time modalities and disruptive behavior disorders in children: a prospective cohort study. *J Child Psychol Psychiatry* 64:125–135. <https://doi.org/10.1111/jcpp.13673>
44. Wartberg L, Kriston L, Zieglermeier M, Lincoln T, Kammerl R (2019) A longitudinal study on psychosocial causes and consequences of internet gaming disorder in adolescence. *Psychol Med* 49:287–294. <https://doi.org/10.1017/S003329171800082X>
45. Swing EL, Gentile DA, Anderson CA, Walsh DA (2010) Television and video game exposure and the development of attention problems. *Pediatrics* 126:214–221. <https://doi.org/10.1542/peds.2009-1508>
46. Azur MJ, Stuart EA, Frangakis C, Leaf PJ (2011) Multiple imputation by chained equations: what is it and how does it work? *Int J Methods Psychiatr Res* 20:40–49. <https://doi.org/10.1002/mpr.329>
47. R Core Team (2023) R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>
48. Hu LT, Bentler PM (1999) Cutoff criteria for fit indexes in covariance structure analysis: conventional criteria versus new alternatives. *Struct Equ Model* 6:1–55. <https://doi.org/10.1080/10705519909540118>
49. Song K, Zhang JL, Zhou N, Fu Y, Zou B, Xu LX, Wang Z, Li X, Zhao Y, Potenza M, Fang X, Zhang JT (2023) Youth screen media activity patterns and associations with behavioral developmental measures and resting-state brain functional connectivity. *J Am Acad Child Adolesc Psychiatry* 62:1051–1063. <https://doi.org/10.1016/j.jaac.2023.02.014>
50. Rigler T, Manor I, Kalansky A, Shorer Z, Noyman I, Sadaka Y (2016) New DSM-5 criteria for ADHD - does it matter? *Compr Psychiatry* 68:56–59. <https://doi.org/10.1016/j.comppsych.2016.03.008>
51. Twenge JM, Martin GN (2020) Gender differences in associations between digital media use and psychological well-being: evidence from three large datasets. *J Adolesc* 79:91–102. <https://doi.org/10.1016/j.adolescence.2019.12.018>

52. Fisher JT, Hopp FR, Chen Y, Weber R (2023) Uncovering the structure of media multitasking and attention problems using network analytic techniques. *Comput Hum Behav* 147:107829. <https://doi.org/10.1016/j.chb.2023.107829>
53. Roberts JA, David ME (2022) Instagram and TikTok flow states and their association with psychological well-being. *Cyberpsychol Behav Soc Netw* 26:80–89. <https://doi.org/10.1089/cyber.2022.0117>
54. SM Perth (Social Media Perth) Instagram Statistics for 2024: Facts and Figures. <https://www.smp Perth.com/resources/instagram/instagram-statistics>
55. Lin X, Su W, Potenza MN (2018) Development of an online and offline integration hypothesis for healthy internet use: theory and preliminary evidence. *Front Psychol* 9:492. <https://doi.org/10.3389/fpsyg.2018.00492>

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.